

Radiation Pressure Instability (RPI)

what to expect when you turn on the lights

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3rd PFITS+
SETI Institute

Radiation Pressure Instability (RPI) — Sculpting With Light and Shadows

Star

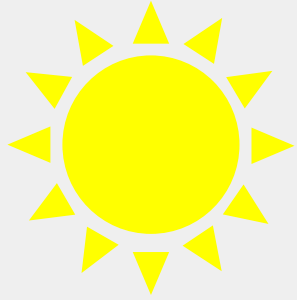


Dust Disk (no gas for now)



$$\Omega_0^2 = \frac{GM}{r^3}$$

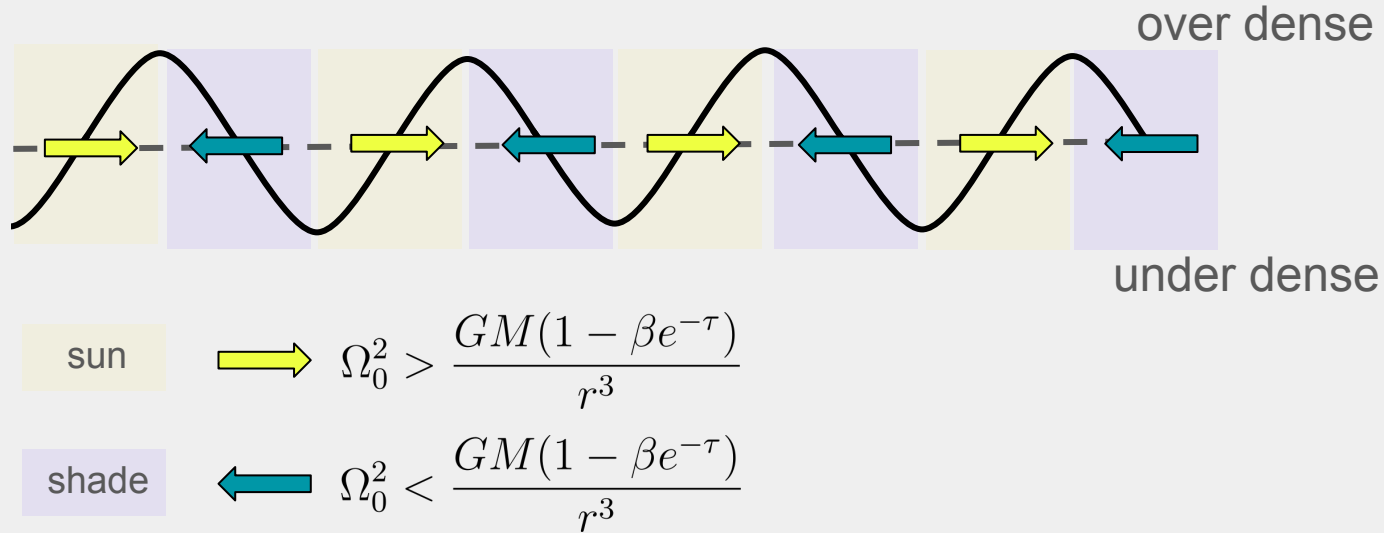
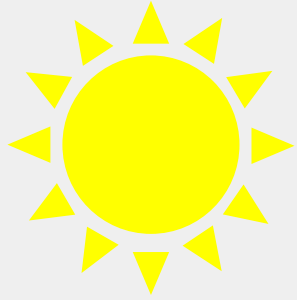
Radiation Pressure Instability (RPI) — Sculpting With Light and Shadows



$$\Omega_0^2 = \frac{GM(1 - \beta e^{-\tau})}{r^3}$$

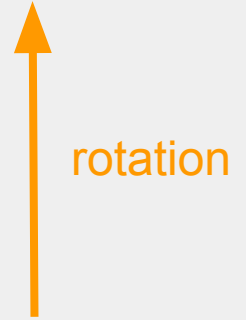
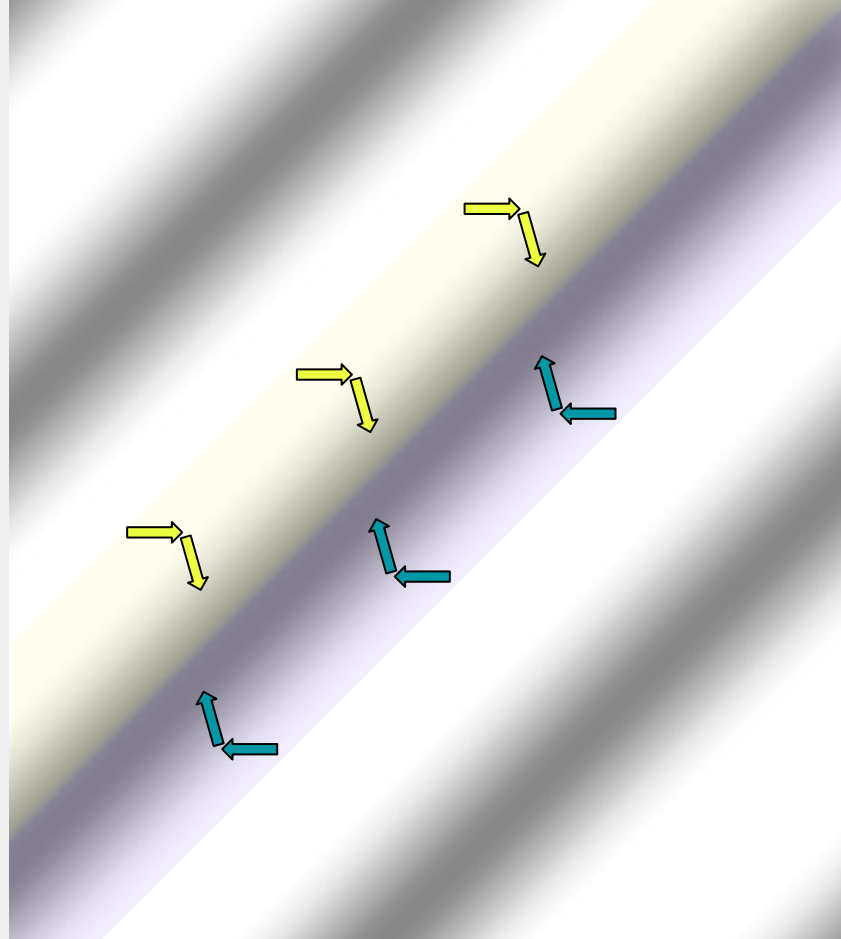
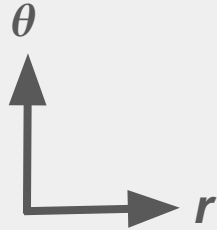
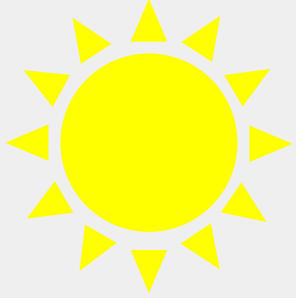
where $\beta = \frac{\kappa_{abs} L}{4\pi G M c}$ and $\tau = \int_0^r \kappa_{opa} \frac{\Sigma}{\sqrt{2\pi h}} dr'$

Radiation Pressure Instability (RPI) — Sculpting With Light and Shadows



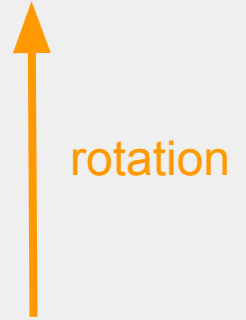
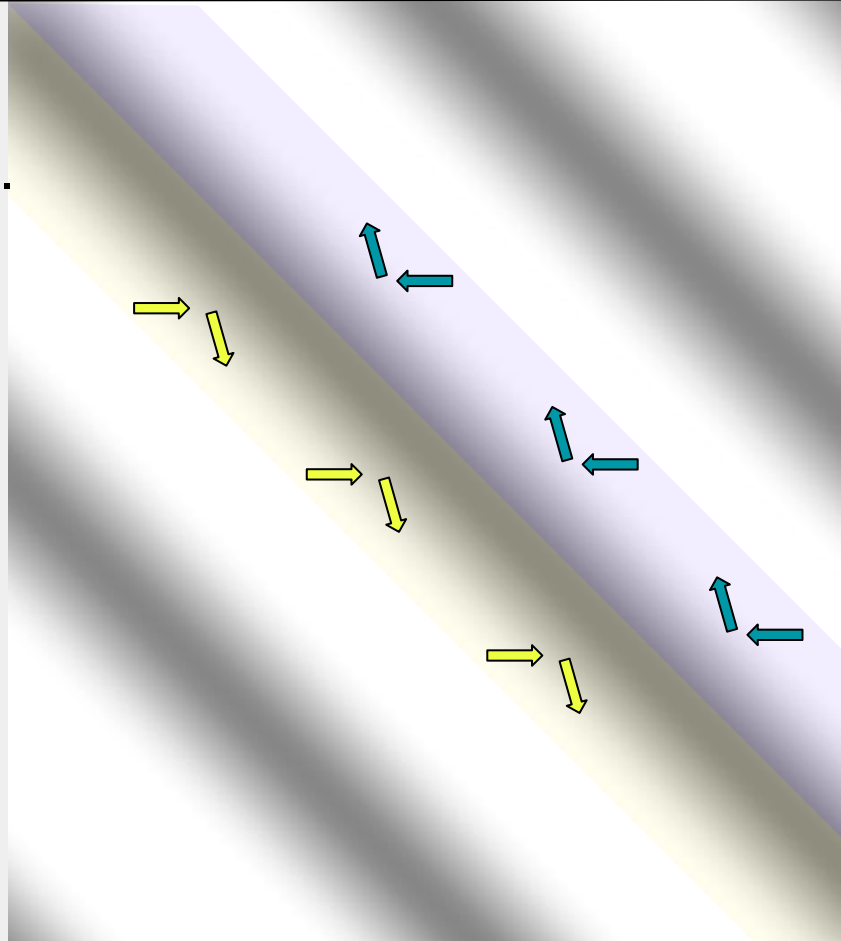
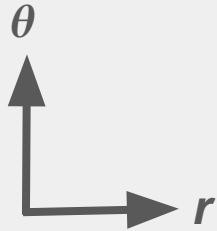
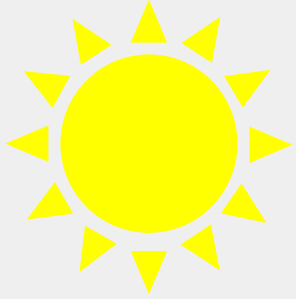
Radiation Pressure Instability (RPI) — Sculpting With Light and Shadows

Leading waves
are unstable.

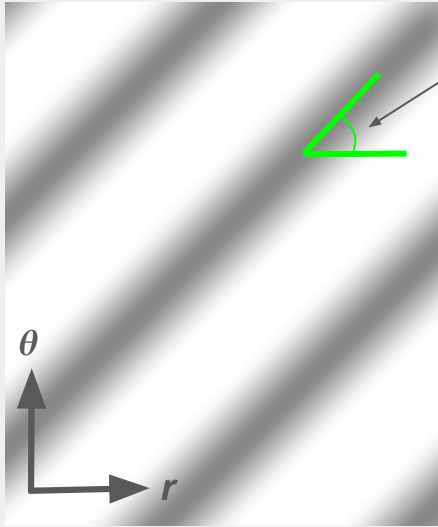


Radiation Pressure Instability (RPI) — Sculpting With Light and Shadows

Trailing waves
are (usually) stable.



Radiation Pressure Instability (RPI) — Sculpting With Light and Shadows



$$\tan^{-1} \left(\frac{\text{radial wavenumber}}{\text{azimuthal wavenumber}} \right) = \tan^{-1} \left(\frac{kr}{m} \right)$$

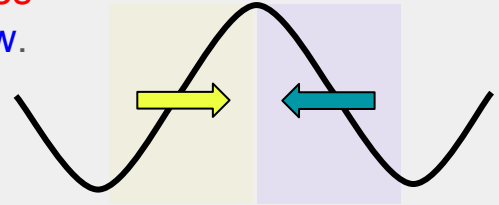
A “quality” factor that measures the potential for radiation pressure to change.

Higher if the local disk material **receives more radiation** and **casts more shadow**.

$k < 0$: leading waves

$k > 0$: trailing waves

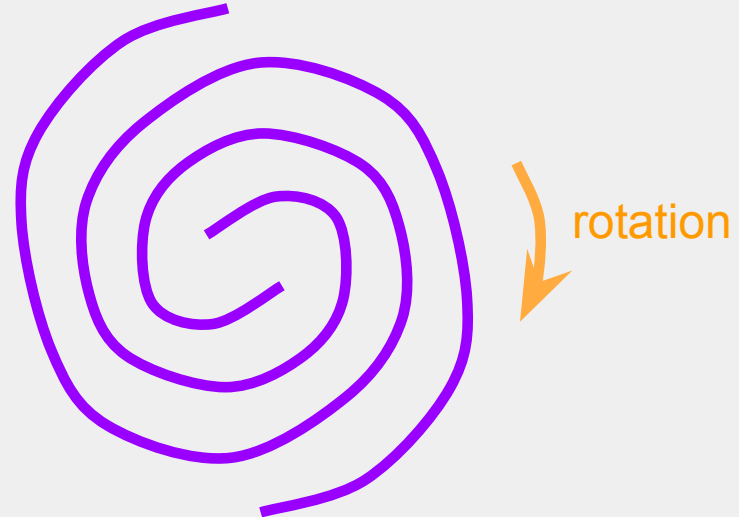
$$Q_{\beta} = \frac{\beta e^{-\tau} \frac{d\tau}{d \ln r}}{1 - \beta e^{-\tau}}$$



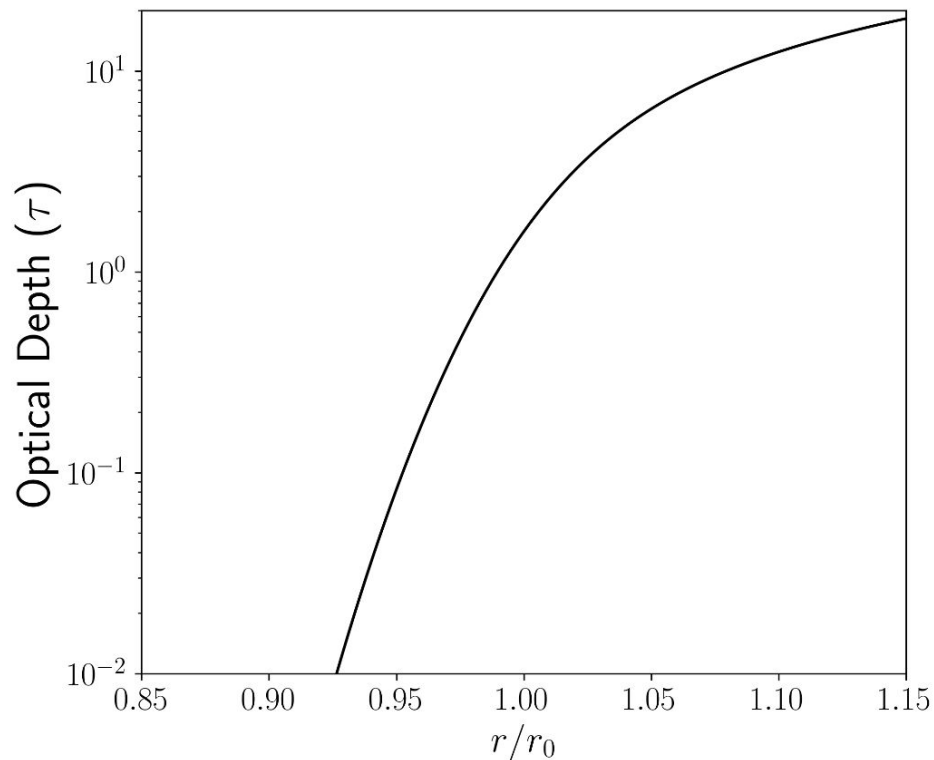
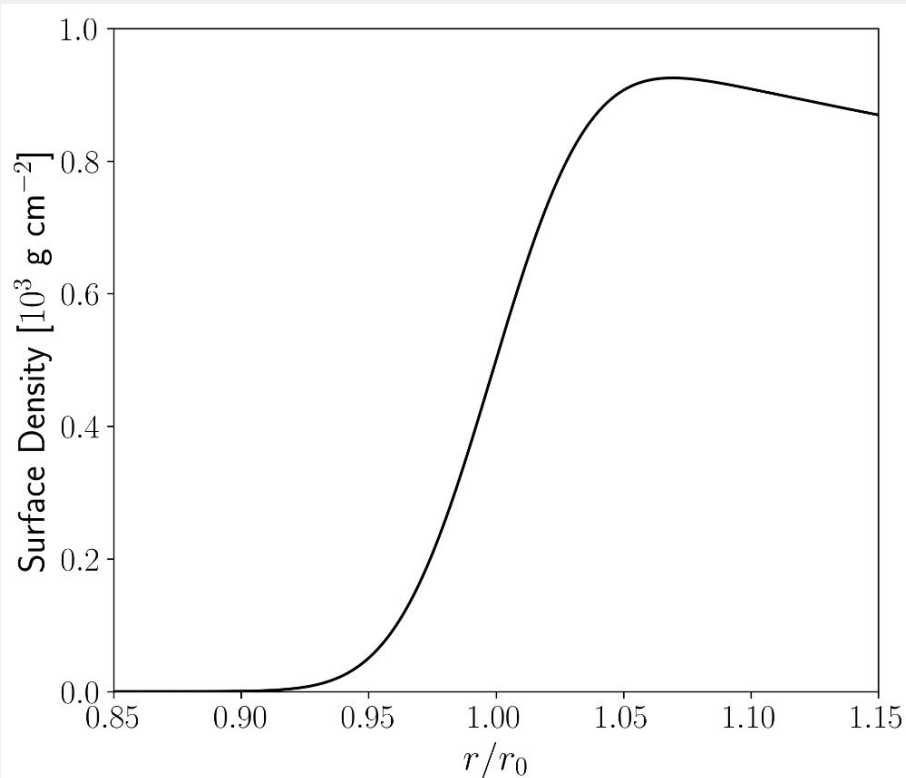
Radiation Pressure Instability (RPI) — Direction of Shear

Leading waves go against the direction of Keplerian shear.

But how Keplerian are these disks?

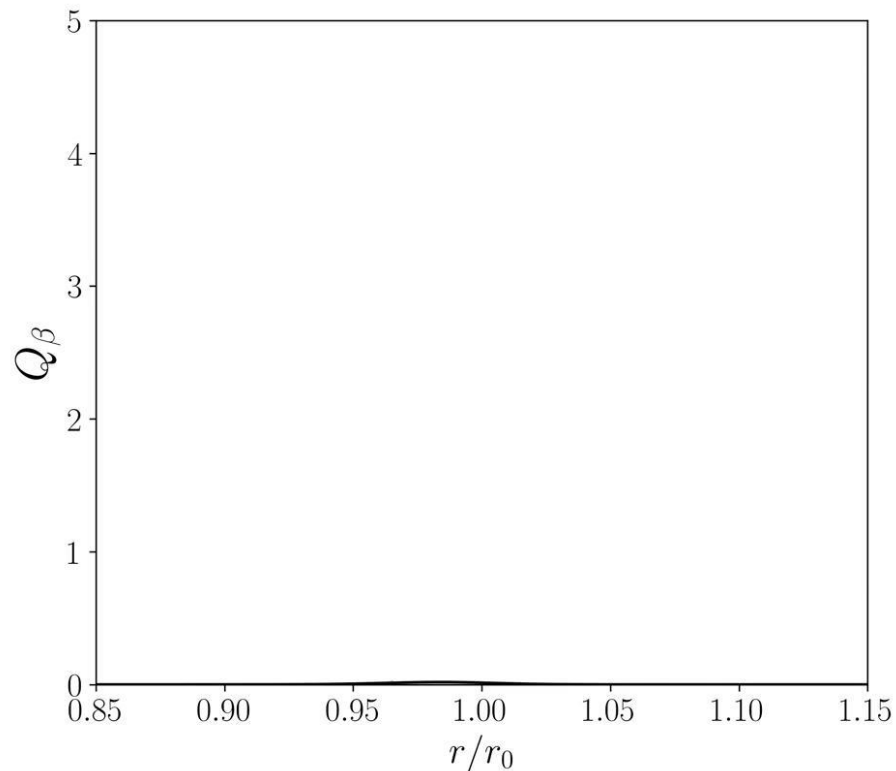
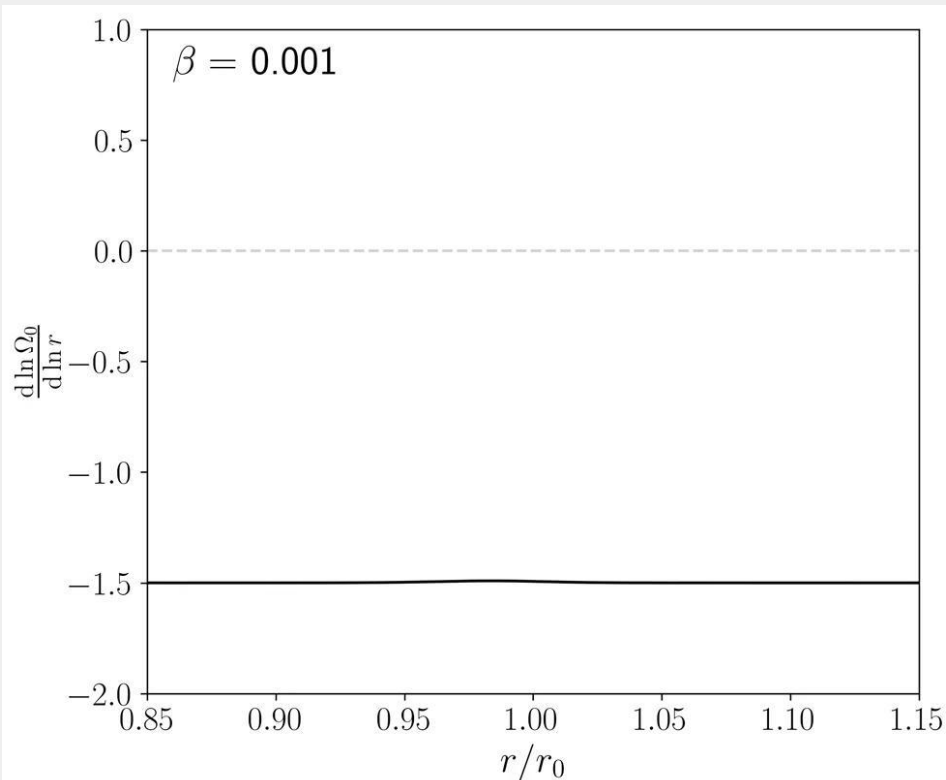


Radiation Pressure Instability (RPI) — Direction of Shear



Radiation Pressure Instability (RPI) — Direction of Shear

Shear direction reverses if $Q_\beta > 3$, supporting the non-linear growth of leading waves.



Radiation Pressure Instability (RPI) — Instability at the Photosphere

We derived a criterion for instability for dust disks:

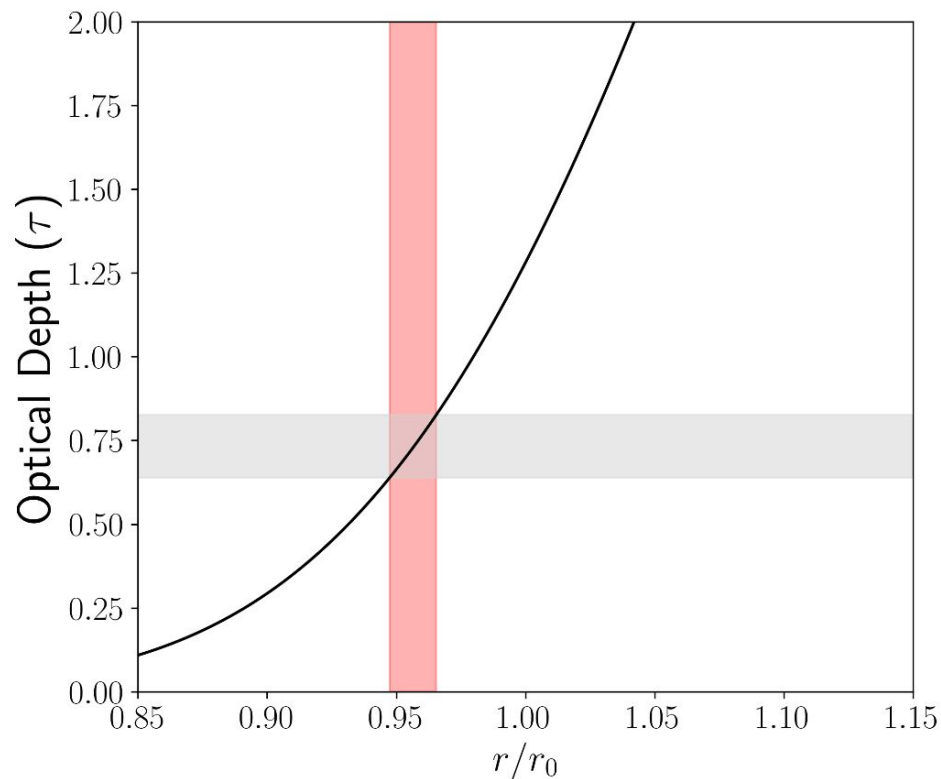
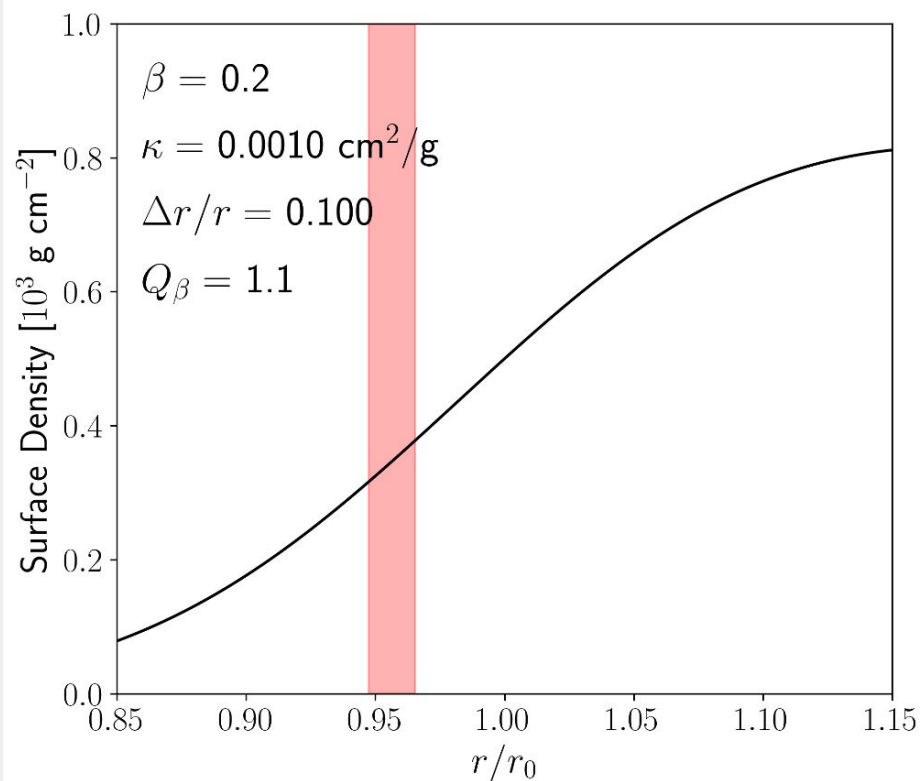
$$2 > \frac{d\tau}{d \ln r} - \frac{d \ln \Sigma}{d \ln r} > -1$$

This essentially refers to the photosphere ($\tau \sim 1$)

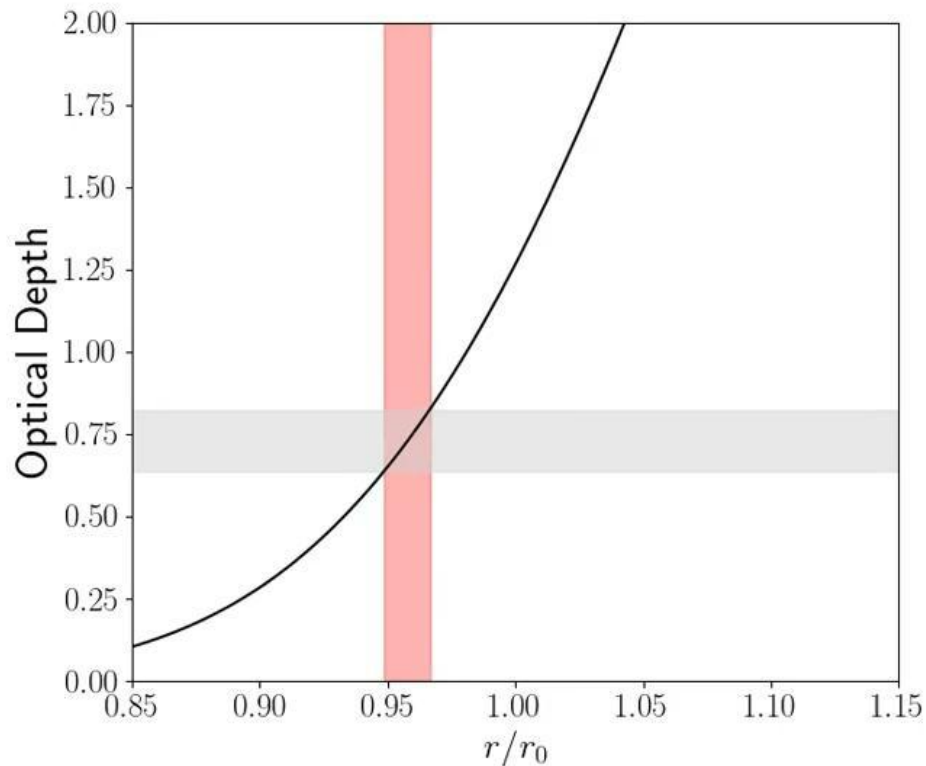
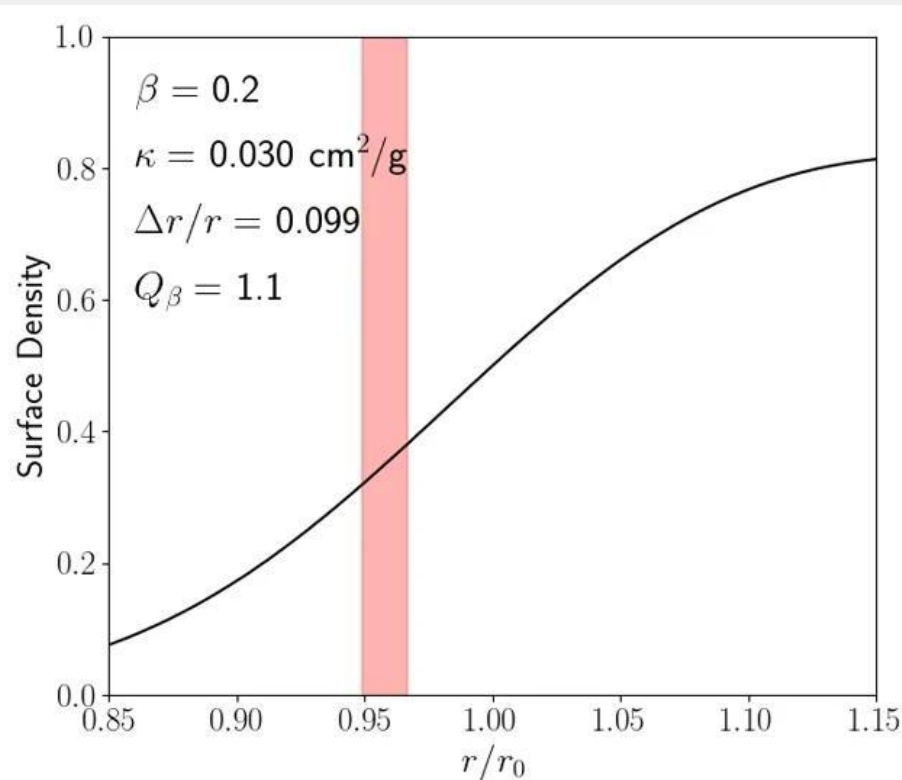
$$2 \frac{\Delta r}{r} \gtrsim \tau - 1 \gtrsim -\frac{\Delta r}{r}$$

where Δr is the width of the photosphere.

Radiation Pressure Instability (RPI) — Instability at the Photosphere



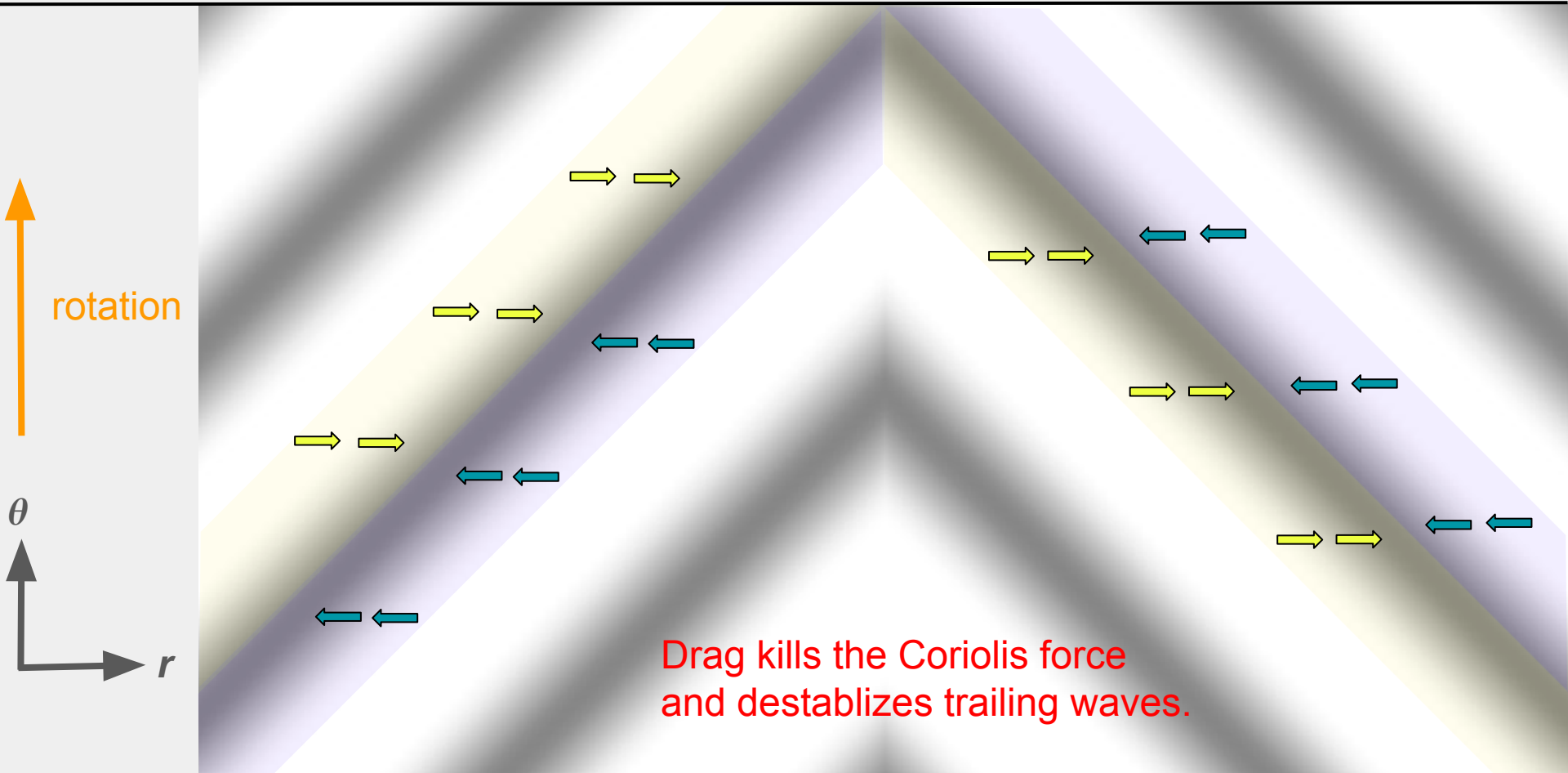
Radiation Pressure Instability (RPI) — Instability at the Photosphere



Radiation Pressure Instability (RPI) — Instability at the Photosphere

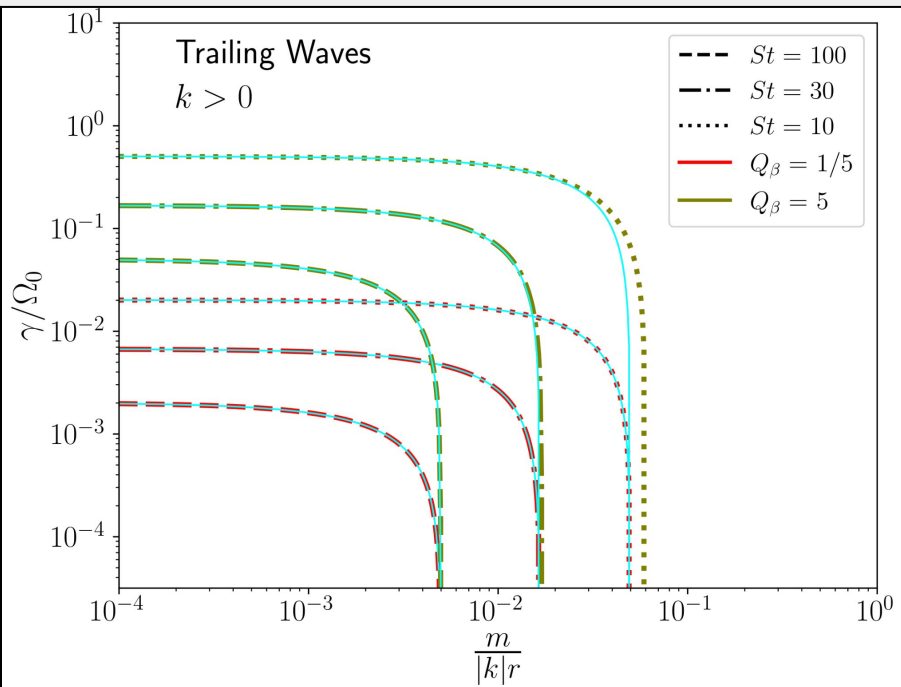
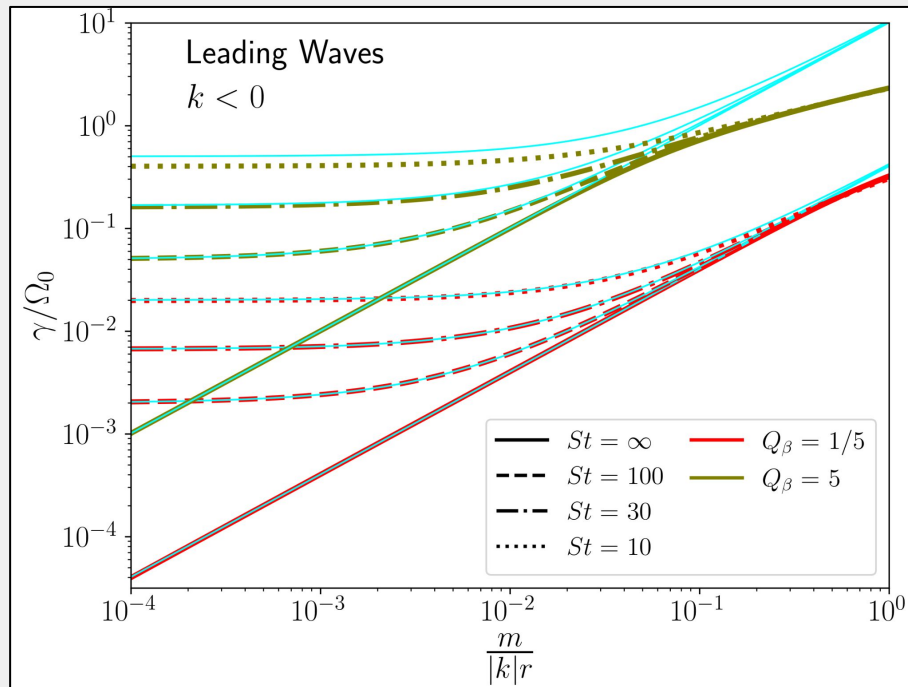
What happens when there is gas?

Radiation Pressure Instability (RPI) — Gas Drag Promotes Instability



Radiation Pressure Instability (RPI) — Gas Drag Promotes Instability

$$\frac{\gamma}{\Omega_0} = \left(-2 \frac{m}{kr} + \frac{1}{St} \right) Q_\beta \quad ; \quad St \gg 1 \quad (\text{weak drag limit})$$



Radiation Pressure Instability (RPI) — Gas Drag Promotes Instability

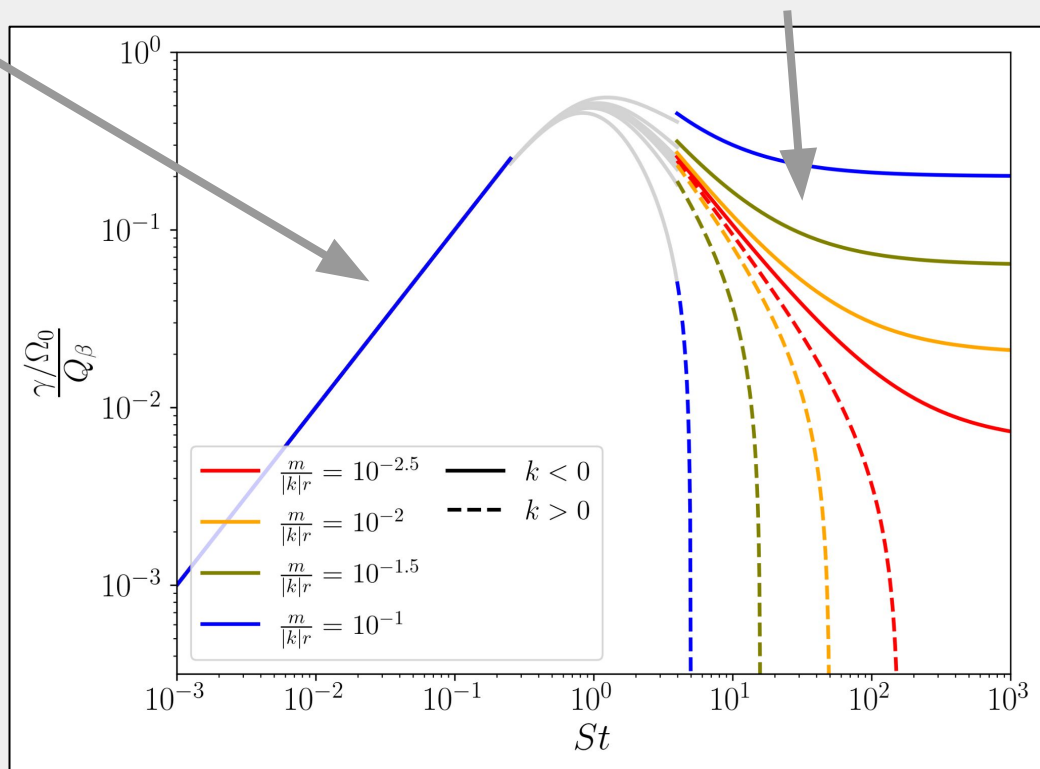
$$\frac{\gamma}{\Omega_0} = St\beta e^{-\tau} \left(\frac{d\tau}{d \ln r} - \frac{1}{2} \right) \sim St Q_\beta \quad ; \quad St \ll 1$$

$$\frac{\gamma}{\Omega_0} = \left(-2 \frac{m}{kr} + \frac{1}{St} \right) Q_\beta \quad ; \quad St \gg 1$$

(strong drag limit)

Growth rate becomes independent of m and k .

All perturbations are unstable at the same rate.



Radiation Pressure Instability (RPI) — Gas Drag Promotes Instability

$$\frac{\gamma}{\Omega_0} = St\beta e^{-\tau} \left(\frac{d\tau}{d\ln r} - \frac{1}{2} \right) \sim StQ_\beta \quad ; \quad St \ll 1$$

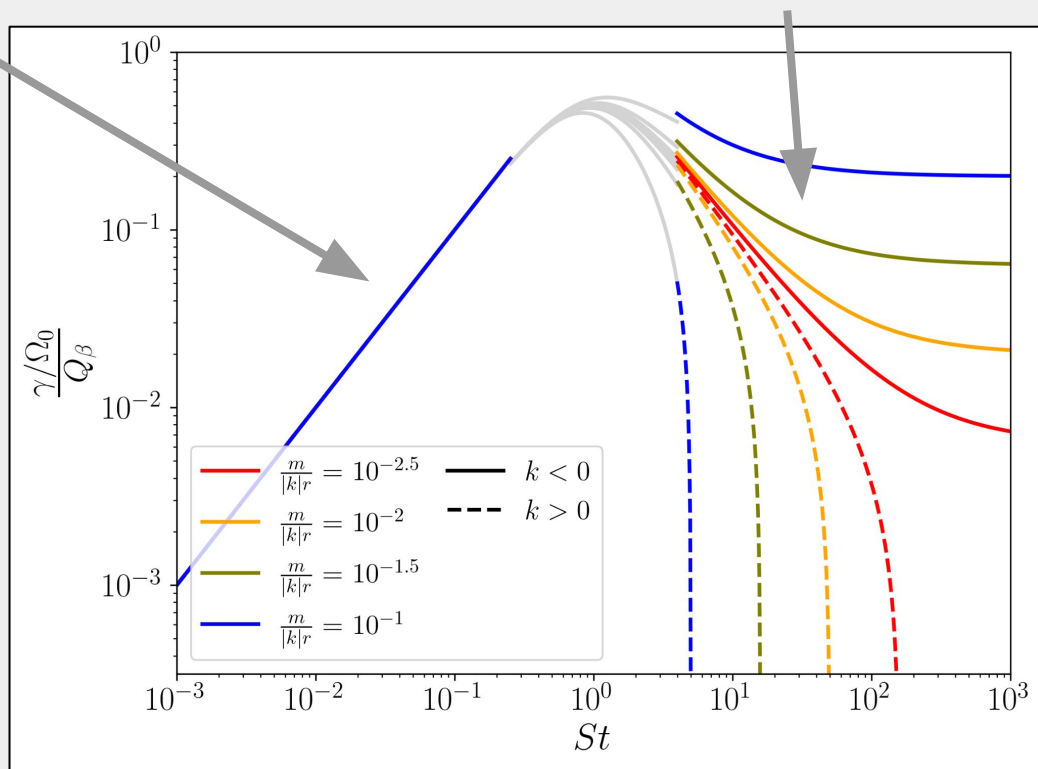
$$\frac{\gamma}{\Omega_0} = \left(-2\frac{m}{kr} + \frac{1}{St} \right) Q_\beta \quad ; \quad St \gg 1$$

(strong drag limit)

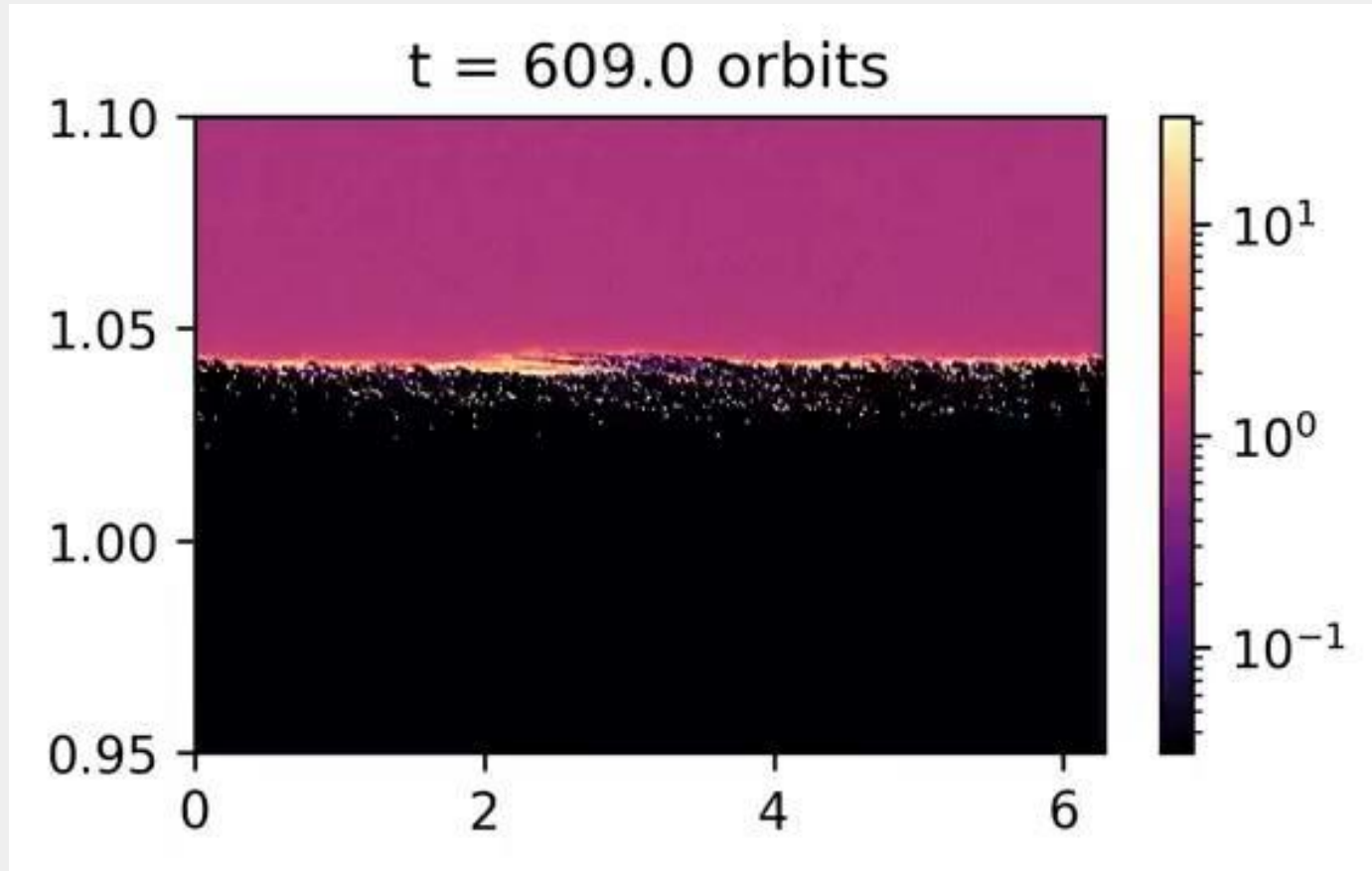
Criterion for instability also relaxes to:

$$\frac{d\tau}{d\ln r} - \frac{d\ln \Sigma}{d\ln r} > \frac{1}{2} \longrightarrow \tau \gtrsim 1$$

But the photosphere is still the most unstable because the growth rate scales with Q_β .

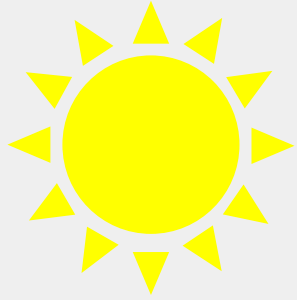


Radiation Pressure Instability (RPI) — Outcome of the Instability



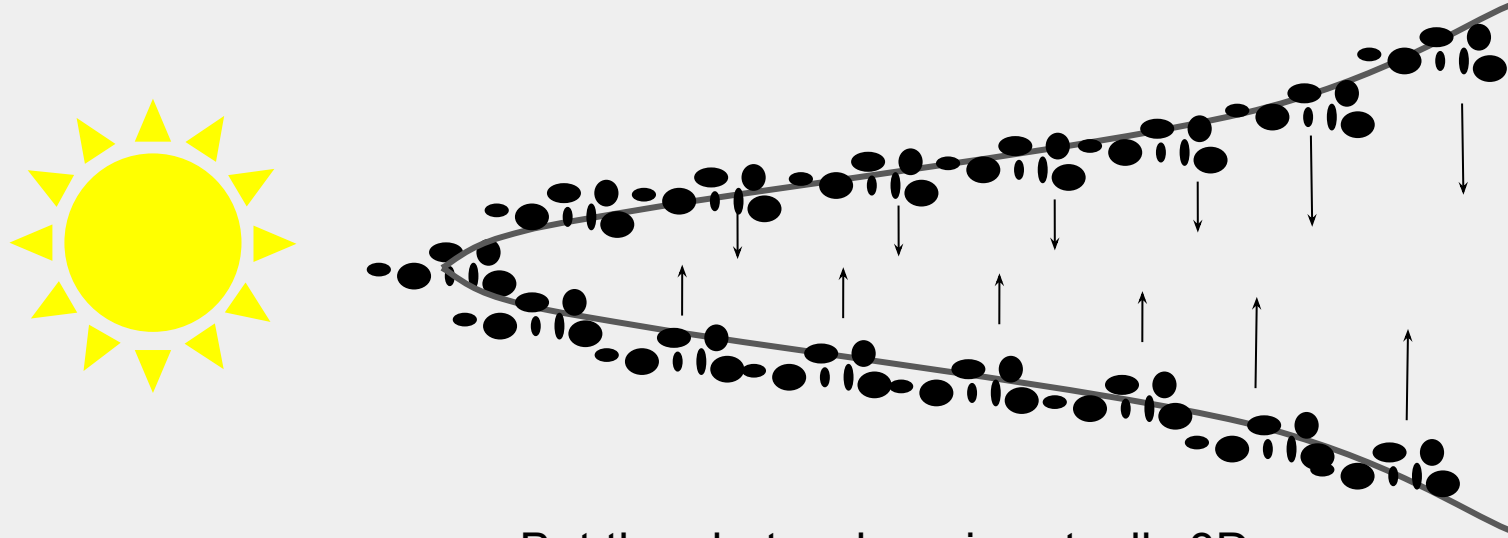
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(Bi+ 2022)

Radiation Pressure Instability (RPI) — Disk Edges vs Disk Surfaces



In 2D, we find dust clumps at disk inner edges.

Radiation Pressure Instability (RPI) — Disk Edges vs Disk Surfaces



But the photosphere is actually 3D.
Pebbles ($St \sim 1$) can form on disk surfaces and rain down to the midplane.

Radiation Pressure Instability (RPI)

- Disk photospheres are unstable to radiation pressure.
 - Uneven disk edges/surfaces should lead to dust emission variability.
 - May explain some IR variability in protoplanetary disks (e.g., Sameshima+ 2026, Thanawuth+ 2026) and white dwarf debris disks (e.g., Noor+ 2025)
- The instability forms compact, high-density dust clumps that should promote grain growth.
- A 3D version of this instability can accelerate global grain growth.